

IN1006 Systems Architecture

Tutorial 3: Logic Gates and Circuits

Exercises

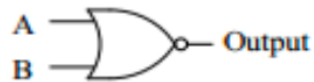
1. Identify each of these logic gates by name and complete their respective truth tables



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1



A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1



A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0



A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0



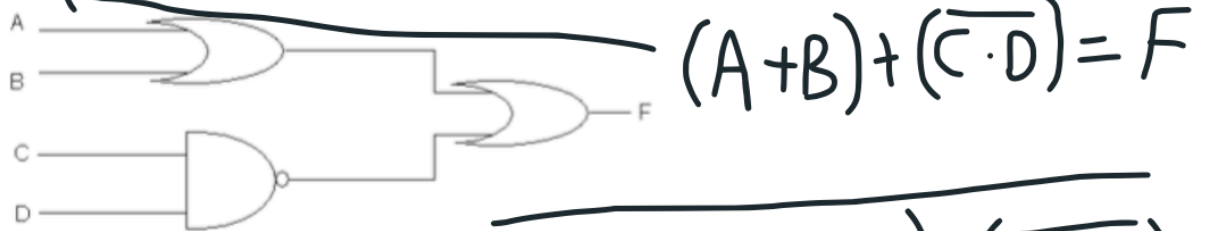
A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

2. Convert the following circuits directly to Boolean expressions:

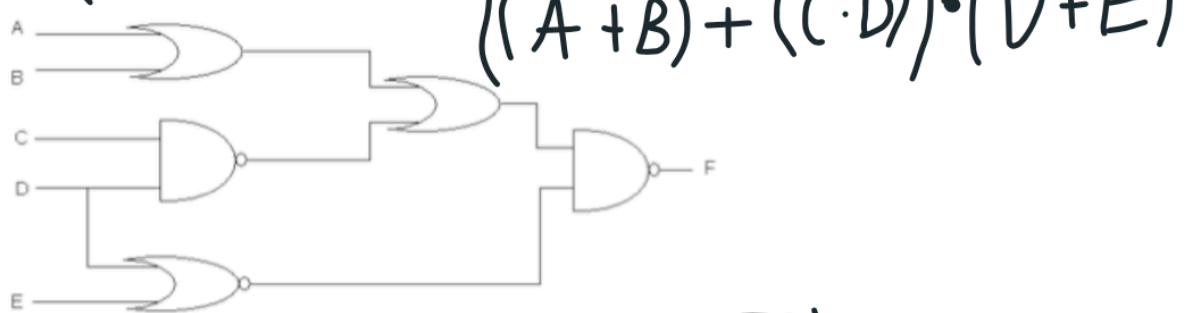
(a).



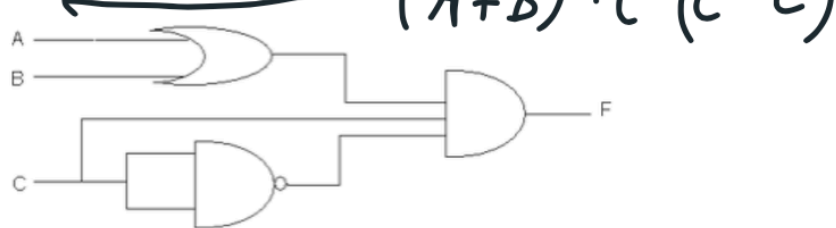
(b).



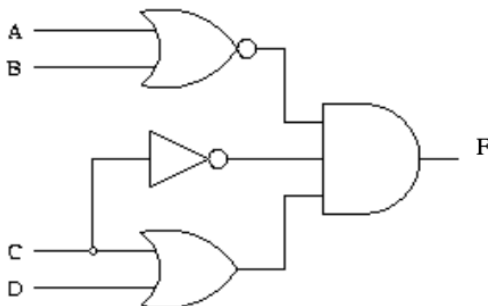
(c).



(d).



(e).



$$(\overline{A} + B) \cdot \overline{C} \cdot (C + D)$$

3. Draw logic diagrams using only AND, OR, and NOT gates to implement the following Boolean expressions. In each case draw the diagrams directly from the expressions and do not attempt to simplify them first.

(a) $F = \bar{A} \cdot B + A \cdot \bar{B}$

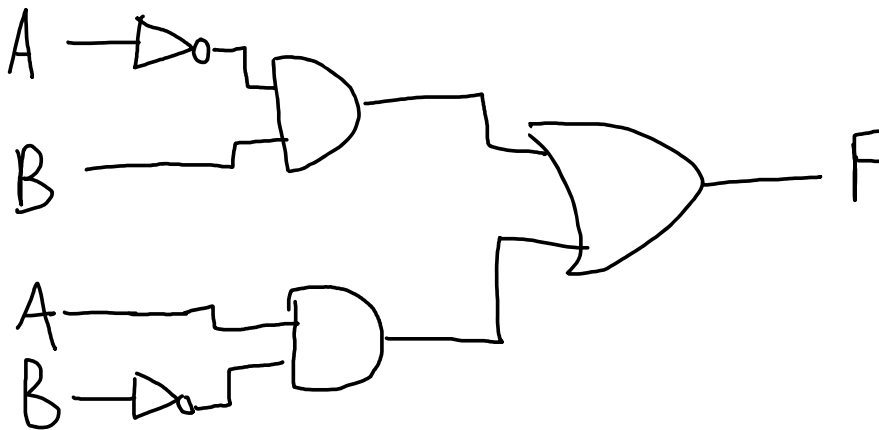
(b) $F = (A + B + C)(A \cdot B + A \cdot C)$

(c) $F = (A + \bar{C})(A + B \cdot \bar{D})$

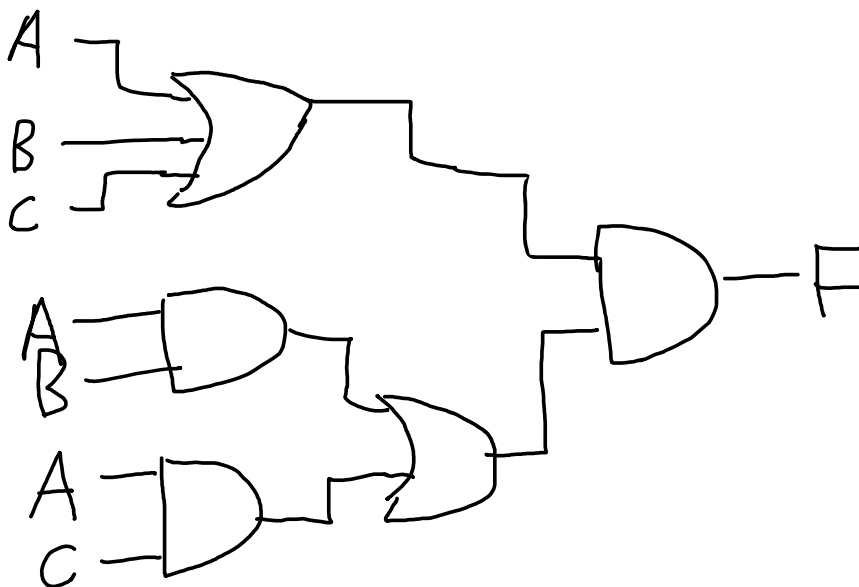
(d) $F = \bar{A} + \bar{C} \cdot A + B \cdot \bar{D}$

(e) $F = (A \cdot \bar{B} + \bar{A} \cdot B + A \cdot C)(\bar{A} \cdot \bar{B} + \bar{A} \cdot \bar{B} + A \cdot \bar{C})$

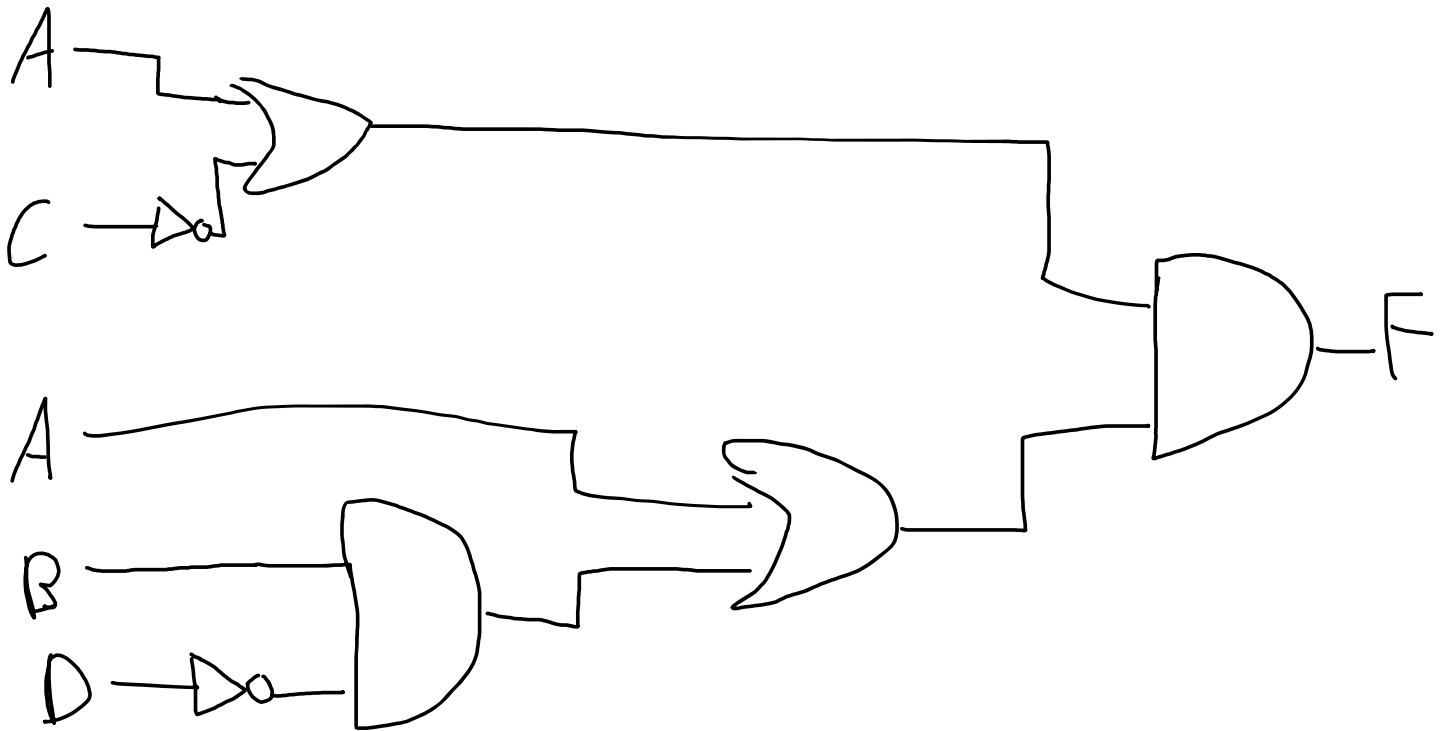
a)



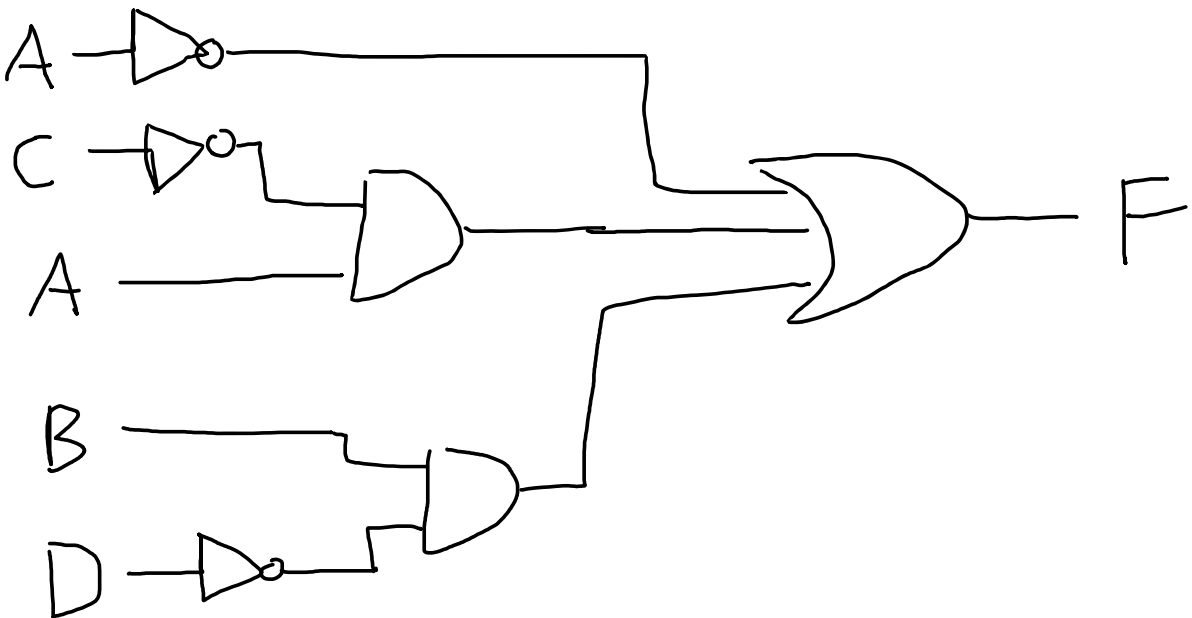
b)



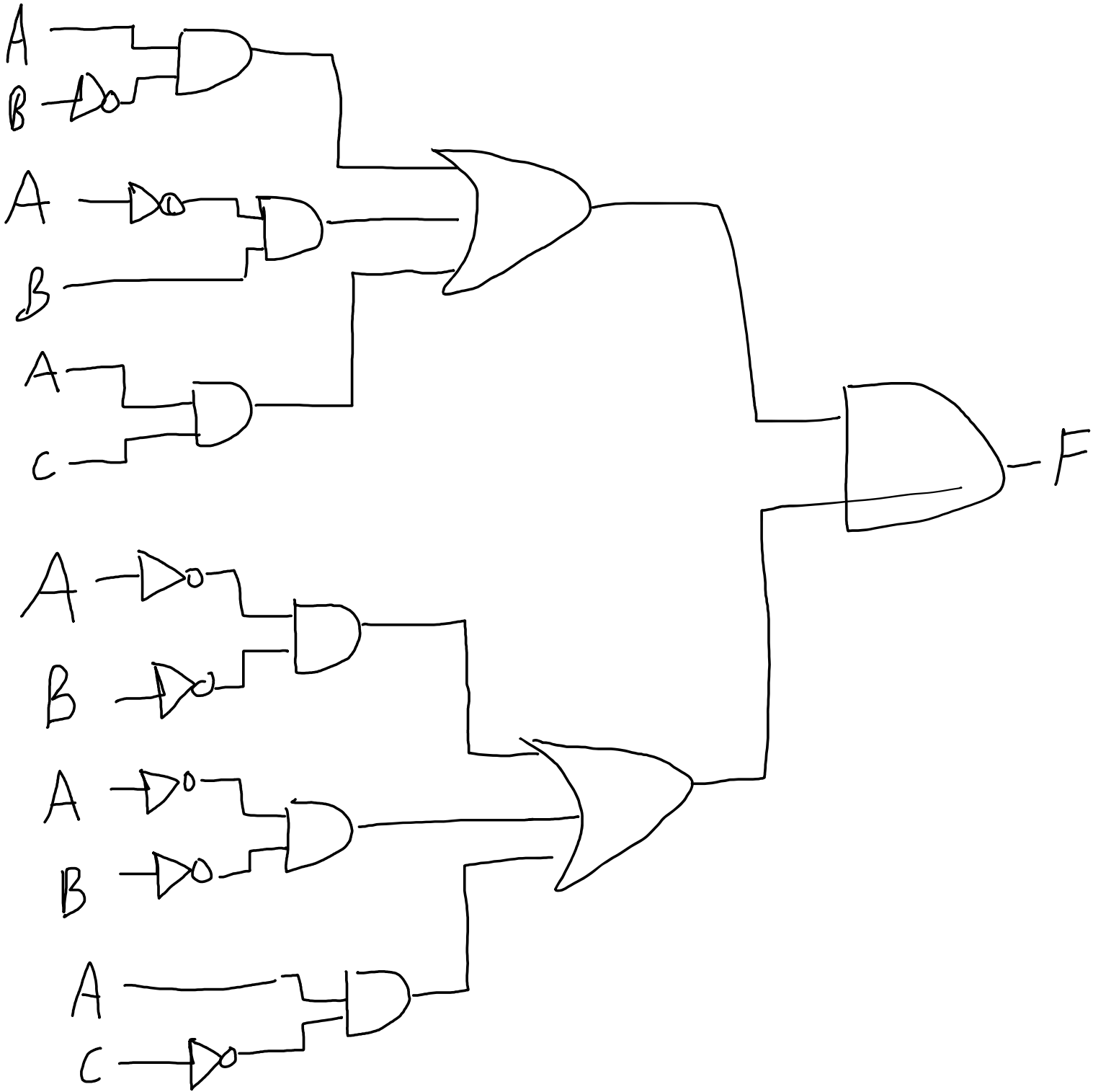
c)



d)



e)



4. Which of the expressions above are in sum-of-products form? How would you obtain a sum-of-products expression from a truth table?

5. Translate this Boolean expressions into its equivalent logic gate circuit:

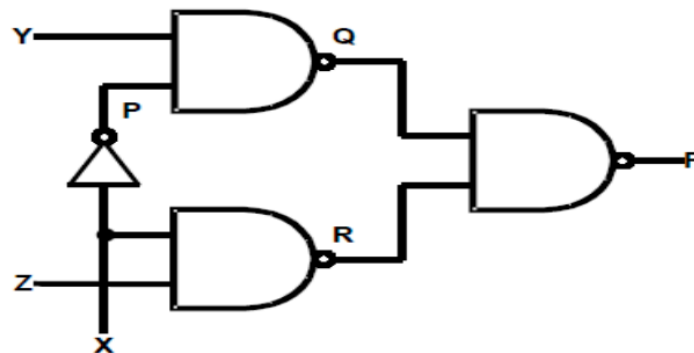
$$F = (A + B + C)(A \cdot B + A \cdot C)$$

6. Create the truth tables out of the following expressions:

(a) $F = \bar{A} \cdot B + A \cdot \bar{B}$

(b) $F = (A + B + C)(A \cdot B + A \cdot C)$

7. Draw the truth table for this logic circuit



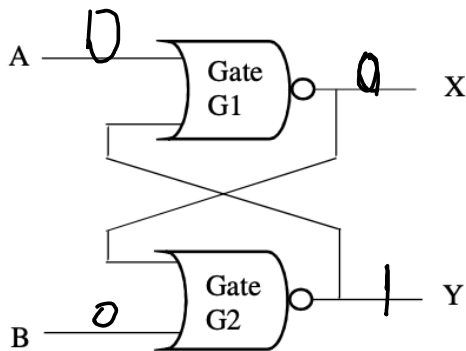
4. The sum of products form takes each set of conditions that return true and represents them as the value of each variable. A and D are in sum of products for.

10. A sequential circuit is one where the output depends not only on the current input but the previous inputs. Combinational circuits do not remember their previous inputs and their outputs are a function of the current inputs.

11. A clocked SR flip flop is attached to a clock.

12. D flip flops create registers

13. Assume you are given the following R-S flip flop with inputs A and B and outputs X and Y.



- a. Assume its initial state is $A=0$, $B=0$, $X=0$ and $Y=1$. What is the effect of setting input B to 1?

X is set to 1 and Y to 0

- b. Assume its initial state is $A=0$, $B=1$, $X=1$ and $Y=0$. What is the effect of setting input B to 0?

No change to X or Y

- c. Assume its initial state is $A=0$, $B=0$, $X=1$ and $Y=0$. What is the effect of setting input A to 1?

X is set to 0 and Y to 1

- d. Assume its initial state is $A=1$, $B=0$, $X=0$ and $Y=1$. What is the effect of setting input A to 0?

No change to X or Y

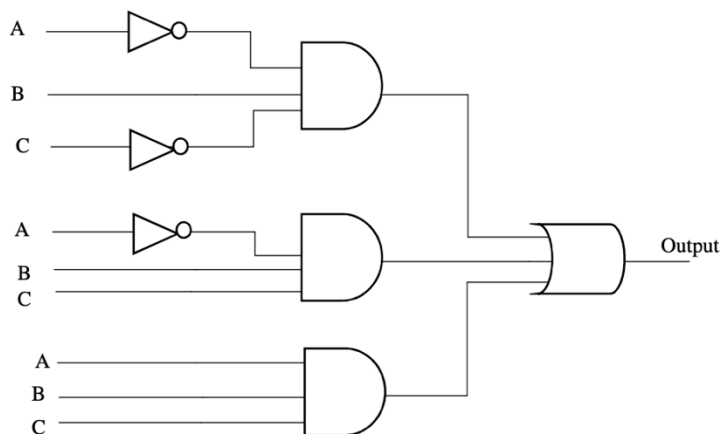
6b)

A	B	C	$A+B+C$	$A \cdot B$	$A \cdot C$	$A \cdot B + A \cdot C$	F
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	1	0	0	0	0
1	0	1	1	0	1	1	1
1	1	0	1	1	0	1	1
1	1	1	1	1	1	1	1

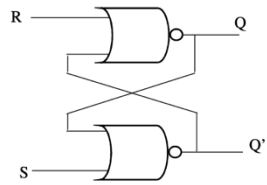
9. Write an SOP expression for this truth table and then draw a gate circuit diagram corresponding to that SOP expression

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

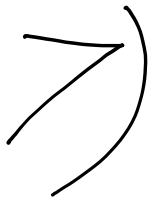
$$\text{Output} = \bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C + A \cdot B \cdot C$$



8. Draw the truth table for the following R-S flip flop

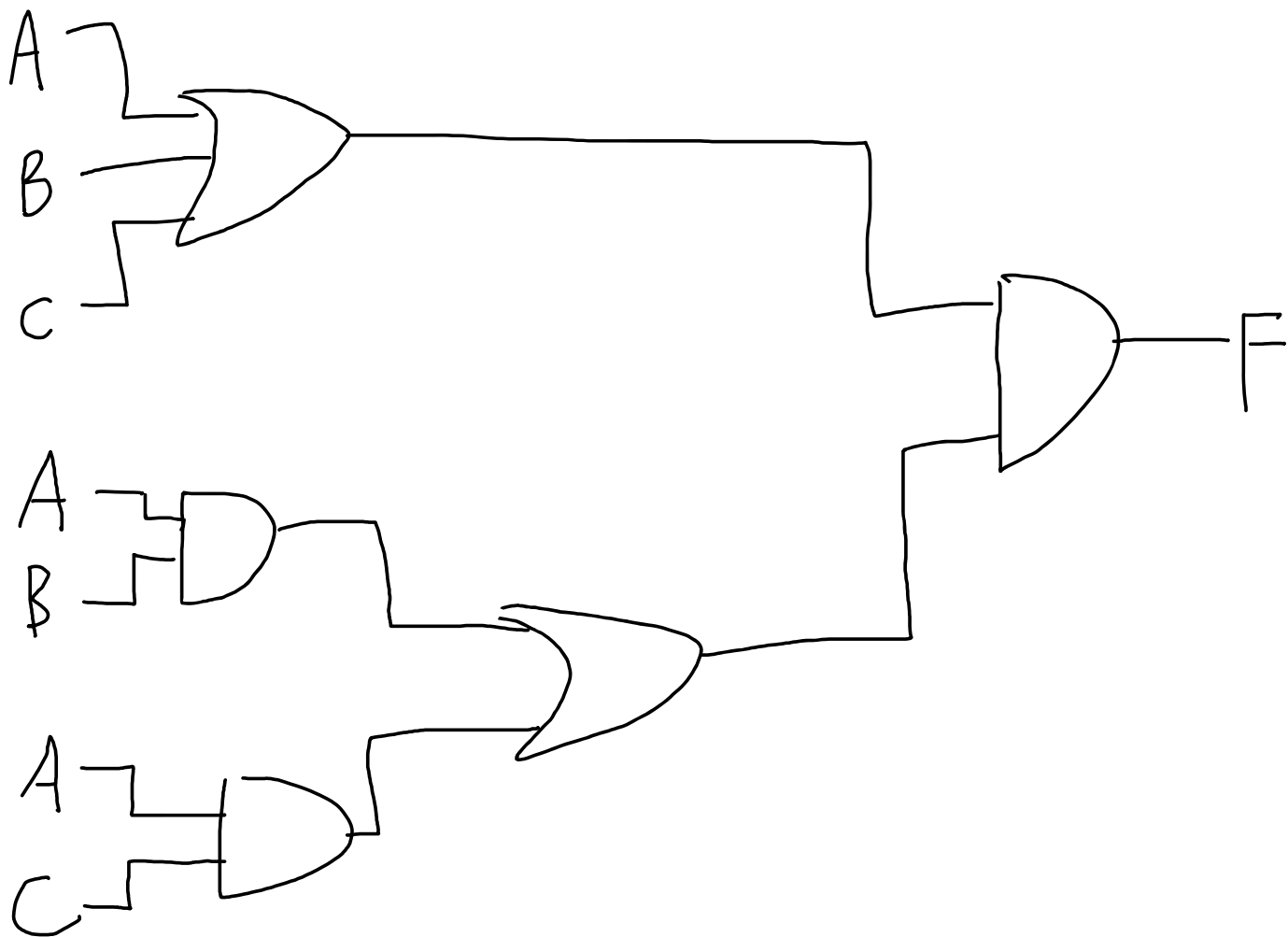


S	R	Q(t)	Q(t + 1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	undefined
1	1	1	undefined



X	Y	Z	P	Q	R	F
0	0	0	1	1	1	0
0	0	1	1	1	1	0
0	1	0	1	0	1	1
0	1	1	0	0	1	0
1	0	0	0	1	0	1
1	1	0	0	1	1	0
1	1	1	0	1	0	1

5)



6)

a)

A	B	$\bar{A}$	$\bar{B}$	$\bar{A} \cdot B$	$A \cdot \bar{B}$	$\bar{A} \cdot B + A \cdot \bar{B}$
0	0	1	1	0	0	0
0	1	1	0	1	0	1
1	0	0	1	0	1	1
1	1	0	0	0	0	0